

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Integrated Experiments

Weightage: 40%

QUESTIONS: Total Marks: 30

Experiment 1: Responsive Layout Design

Suppose that a retail website is not responsive across mobile and tablet devices, causing misalignment of content and improper spacing.

(a) Analyze the possible reasons for lack of responsiveness in the existing layout.

A responsive website automatically adjusts its layout according to the screen size of the device. If a retail website is not responsive, users may experience overlapping text, improper spacing, horizontal scrolling, and poor readability.

Possible Reasons:

1. Fixed Width Layout
 - The webpage uses fixed widths such as `width:1200px` instead of flexible widths (`100%`, `auto`, `%`).
 - Fixed layouts cannot adapt to different screen sizes.
2. No Viewport Meta Tag
3. Without the viewport tag, mobile browsers display the webpage as a desktop.

(b) Design a responsive webpage layout using CSS Flexbox or Grid, ensuring proper alignment of header, navigation, content, and footer.

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Responsive Layout</title>
<link rel="stylesheet" href="style.css">
</head>

<body>

<header>
<h1>Online Shopping</h1>
</header>

<nav>
<a href="#">Home</a>
<a href="#">Products</a>
<a href="#">Offers</a>
<a href="#">Contact</a>
</nav>

<div class="container">

<section class="content">
<h2>Welcome</h2>
<p>This is the main content area.</p>
</section>

<aside class="sidebar">
<h3>Categories</h3>
<ul>
<li>Electronics</li>
<li>Fashion</li>
<li>Books</li>
</ul>
</aside>

</div>

<footer>
```

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</footer>

</body>

</html>

(c) Demonstrate how the layout adapts to different screen sizes (mobile, tablet, desktop) using media queries.

```
@media screen and (max-width:768px){
```

```
.container{  
flex-direction:column;  
}
```

```
nav{  
flex-direction:column;  
}
```

```
nav a{  
text-align:center;  
border-bottom:1px solid white;  
}
```

```
}
```

(d) Implement spacing, alignment, and distribution techniques to improve visual consistency.

```
.container{  
display:flex;  
justify-content:space-between;  
align-items:flex-start;  
gap:20px;  
padding:20px;  
}
```

```
.content,  
.sidebar{  
border-radius:10px;  
box-shadow:0 0 10px gray;  
padding:20px;  
}
```

(e) Evaluate the effectiveness of the responsive design and suggest further improvements.

Evaluation

The responsive webpage works efficiently on desktop, tablet, and mobile devices.

Advantages

- Supports multiple screen sizes.
- Eliminates horizontal scrolling.
- Proper alignment of all sections.
- Improves readability.
- Enhances user experience.
- Easy to maintain using Flexbox.

Experiment 2: Form Validation using JavaScript

Suppose that a registration form accepts invalid email and phone number inputs, leading to incorrect data submission.

(a) Analyze the issues in the existing form validation mechanism.

Invalid email accepted.

Phone number accepts letters.

Empty fields submitted.

No error messages.

Poor user experience.

(b) Design input fields for email and phone number with appropriate HTML attributes.

<form>

Email:

<input type="email" id="email" required>

Phone:

<input type="text" id="phone" required maxlength="10">

<p id="msg"></p>

<button onclick="return validate()">Submit</button>

</form>

(c) Implement JavaScript validation using Regular Expressions for both email and phone number.

```

function validate(){

var email=document.getElementById("email").value;
var phone=document.getElementById("phone").value;

var emailPattern=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;
var phonePattern=/^[0-9]{10}$/;

if(!emailPattern.test(email)){
alert("Invalid Email");
return false;
}

if(!phonePattern.test(phone)){
alert("Invalid Phone Number");
return false;
}

alert("Validation Successful");
return true;

}

```

(d) Develop a mechanism to display real-time error messages for invalid inputs.

```

document.getElementById("email").onkeyup=function(){

let email=this.value;
let pattern=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;

document.getElementById("msg").innerHTML=
pattern.test(email)?"Valid Email":"Invalid Email";

}

```

(e) Evaluate the validation system and suggest enhancements for better user experience and security.

Advantages

- Prevents invalid input.
- Better user experience.
- Reduces server errors.

Enhancements

- Password validation.
- CAPTCHA.
- Server-side validation.

- Input sanitization.

Experiment 3: CSS Layout Debugging

Suppose that a webpage layout overlaps when the browser window is resized, affecting usability.

(a) Analyze the CSS properties responsible for layout overlapping issues.

Analyze overlapping issues

- Fixed width elements.
- Absolute positioning.
- Missing Flexbox/Grid.
- Overflow hidden.
- Incorrect z-index.

(b) Examine the role of the CSS box model (margin, border, padding, content) in layout design.

CSS Box Model

The box model contains:

- Content
- Padding
- Border
- Margin

Proper use prevents layout overlap.

(c) Debug the layout using appropriate positioning techniques (relative, absolute, flex, grid).

```
.container{  
display:flex;  
flex-wrap:wrap;  
gap:20px;  
}
```

```
.box{  
flex:1;  
min-width:250px;  
padding:20px;  
}
```

(d) Modify the CSS to ensure proper alignment and prevent overlapping during resizing.

```
*{  
box-sizing:border-box;  
}
```

```
.container{  
display:flex;  
flex-wrap:wrap;  
justify-content:center;  
}
```

```
img{  
max-width:100%;  
height:auto;  
}
```

(e) Evaluate the corrected layout and suggest best practices for maintaining consistency.

Result

- No overlapping.
- Responsive layout.
- Better readability.
- Improved usability.

Best Practices

- Use Flexbox/Grid.
- Avoid fixed widths.
- Use media queries.
- Use `box-sizing:border-box`.
- Test on different screen sizes.
- Optimize images.

Code of Conduct:

I K . Rama koti Reddy(192372208) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: